Submission Worksheet

Submission Data

Course: IT202-008-S2026
Assignment: IT202 PHP Challenge Scenarios
Student: Kurt C. (kc784)
Status: Submitted | **Worksheet Progress:** 100+%
Potential Grade: 11.00/10.00 (110.00%)
Received Grade: 0.00/10.00 (0.00%)
Started: 2/23/2026 12:07:46 PM
Updated: 2/23/2026 11:36:24 PM
Grading Link: https://learn.ethereallab.app/assignment/v3/IT202-008-S2026/it202-php-challenge-scenarios/grading/kc784
View Link: https://learn.ethereallab.app/assignment/v3/IT202-008-S2026/it202-php-challenge-scenarios/view/kc784

Instructions

- Overview Link: https://youtu.be/iwA6vyRPhoM (older reference, but should be accurate for now)

1. Ensure you read all instructions and objectives before starting. It's recommended to solve all the requirements first, then fill out the worksheet.
2. Create a new branch from `qa` called `M2-Homework`
    1. `git checkout qa` (ensure proper starting branch)
    2. `git pull origin qa` (ensure history is up to date)
    3. `git checkout -b M2-Homework` (create and switch to branch)
3. Copy the template code from here: GitHub Repository - M2 Homework
    - It includes Scenarios 1-4 and `base.php`. Put all into an `M2` folder or similar inside your `public_html`
    - Immediately record to history
        - [ ] `git add public_html`
        - [ ] `git commit -m "adding M2 HW baseline files"`
        - [ ] `git push origin M2-Homework`
        - [ ] Create a Pull Request from `M2-Homework` to `qa` and keep it open
4. Fill out the below worksheet
    - Each Scenario requires the following as you work
        - [ ] Ensure there's a comment with your UCID, date, and brief summary of how the problem was solved
        - [ ] Initial outline/plan of how you'll solve it via comments (add/commit after this stage)
        - [ ] Code solution (add/commit periodically as needed)
    - If you chose to tackle this problem by problem and fill out the worksheet as you go, you'll likely need to keep merging pull requests into `qa`
5. Once finished, click "Submit and Export"
6. Locally add the generated PDF to a folder of your choosing inside your repository folder (homework folder is fine) and move it to Github
    1. `git add .`
    2. `git commit -m "adding PDF"`

3. `git push origin M2-Homework`
4. On Github merge the pull request from `M2-Homework` to `qa`
5. On Github create a pull request from `qa` to `prod` and immediately merge. (This will trigger the `prod` deploy to make the Render.com prod links work)

7. Upload the same PDF to Canvas
8. Sync Local
    1. `git checkout qa`
    2. `git pull origin qa`

Section #1: (2 pts.) Scenario 1 - Odds

Progress: 100%

≡ Task #1 (2 pts.) - Edit the `printOdds` function to output odd values of the array

Progress: 100%

Details:

- Only make edits where noted via provided comments
- Challenge 1: From each passed in array, print odd values only in a single line separated by commas and a space after each comma (should not have leading or trailing commas)
- Step 1: sketch out plan using comments (include ucid and date)
- Step 2: Add/commit your outline of comments (required for full credit)
- Step 3: Add code to solve the problem (add/commit as needed)

🖼 Part 1:

Progress: 100%

Details:
Two screenshots are expected

1. Snippet of relevant code showing solution (with ucid/date comment present)
2. Full output of executing the program (visit the proper file on Render.com qa after a manual deploy)

```php
$output_result = "";
// Start Solution Edits
// set solution to $output_result variable

// kc784 | 2/18
// iterate through each element + check if divisible by 2
$firstOdd = true;
foreach ($arr as $num) {
    if ($num % 2 != 0) {        // iterates through all elements + checks if divisble by 2
        if (!$firstOdd) {
            $output_result .= ", ";      // separates output by commas
        }
        $output_result .= $num;
        $firstOdd = false;
    } <-- #36-42 if ($num % 2 != 0)
} <-- #35-43 foreach ($arr as $num)

// End Solution Edits
printScenario1Output($output_result);
```

relevant code





program via render.com qa

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Part 2:

Progress: 100%

Details:

- Direct link to the file in the homework related branch from Github (should end in `.php`, , zero credit if it does not)
- Direct link to the file on Render.com Prod (Grab the qa url you're testing, paste it, change "-qa" to "-prod"), zero credit if

URL #1
https://github.com/kurtjcastro/kc784-it202-008-2026/M2-Homework/public_html/M2/scenario1.php

URL
https://github.com/kurtjcastro/kc

URL #2
https://kc784-it202-008-prod.onrender.com/M2/scenario1.php

URL
https://kc784-it202-008-prod.onre

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Part 3:

Progress: 100%

Details:
Briefly explain how your solution works. Describe the logic, steps, or decisions your code makes to reach the result. Do not restate the problem or list requirements.

Your Response:

The code works because it loops through each element in the array and checks each number if they are divisible by 2. If the remainder is not 0, it is added to the output as an odd number where I used a boolean to take care of formatting and after each element is a comma separator.

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Section #2: (2 pts.) Scenario 2 - Sum

Progress: 100%

≡ Task #1 (2 pts.) - Edit the `sumValues` function to sum the array values and present them in a format with exactly two decimal places

Progress: 100%

Details:

- Only make edits where noted via provided comments
- Challenge 1: Sum all the values of the passed in array and assign to the `total` variable
- Challenge 2: Have the sum (total) be represented as a number with exactly 2 decimal places (similar to currency), assign to `modifiedTotal` variable
- Example: 0.1 would be shown as 0.10, 1 would be shown as 1.00, 0.011 as 0.01, etc
- Step 1: sketch out plan using comments (include ucid and date)
- Step 2: Add/commit your outline of comments (required for full credit)
- Step 3: Add code to solve the problem (add/commit as needed)

Part 1:

Progress: 100%

Details:
Two screenshots are expected

1. Snippet of relevant code showing solution (with ucid/date comment)
2. Full output of executing the program (visit the proper file on Render.com qa after a manual deploy)

```php
$total = 0;
// Start Solution Edits
// Solve Challenge 1 here: Sum all values
// kc784 | 2/23
// check elements of array and sum all values then reformat

for ($i = 0; $i < count($arr); $i++) {
    $total += $arr[$i];      // sum of all array elements
}

// Solve Challenge 2 here: Format to 2 decimal places
$modifiedTotal = number_format($total, 2, ".", "");     // formatting by 2 dp

// End Solution Edits
printScenario2Output($total, $modifiedTotal);
echo "</div>";
```

relevant code



program execution vs render.com qa

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Part 2:

Progress: 100%

Details:

- Direct link to the file in the homework related branch from Github (should end in `.php`, , zero credit if it does not)
- Direct link to the file on Render.com Prod (Grab the qa url you're testing, paste it, change "-qa" to "-prod"), zero credit if , zero credit if it's not the direct link to the file or doesn't load

URL #1
https://github.com/kurtjcastro/kc784-it202-008-2026/M2-Homework/public_html/M2/scenario2.php

URL
https://github.com/kurtjcastro/kc

URL #2
https://kc784-it202-008-prod.onrender.com/M2/scenario2.php

URL
https://kc784-it202-008-prod.onre

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Part 3:

Progress: 100%

Details:
Briefly explain how your solution works. Describe the logic, steps, or decisions your code makes to reach the result. Do not restate the problem or list requirements. Additionally, did you notice anything about the math of some of the data? What do you think happened and what's the reason for it?

Your Response:

I used a for loop to iterate through each array by the index and stored all values into $total. After every element is accounted for, I store a different set of display values in modifiedTotal so that the sum is rounded by 2 decimals. I used PHP's number_format to help me round the numbers for the output.

Yes, the math of the data showed really small values like numbers that should be 0 but, for example, are 0.000000001. Since computing algorithms have an issue with float and decimal calculations, it takes the closest approximation, but I still found it easy to reformat by 2 decimal places even though it'll be a hassle down the line.

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Section #3: (2 pts.) Scenario 3 - Conversion

Progress: 100%

☰ Task #1 (2 pts.) - Edit the `bePositive` function to make each value positive, convert it back to the original data type, and set it to the proper slot in the `output` array

Progress: 100%

Details:

- Only make edits where noted via provided comments
- Challenge 1: Make each value positive
- Challenge 2: Convert the values back to their original data type and assign it to the proper slot of the `output` array
- Step 1: sketch out plan using comments (include ucid and date)
- Step 2: Add/commit your outline of comments (required for full credit)
- Step 3: Add code to solve the problem (add/commit as needed)

🖼 Part 1:

Progress: 100%

Details:
Two screenshots are expected

1. Snippet of relevant code showing solution (with ucid/date comment)
2. Full output of executing the program (visit the proper file on Render.com qa after a manual deploy)



relevant code





program execution via render.com qa

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Part 2:

Progress: 100%

Details:

- Direct link to the file in the homework related branch from Github (should end in `.php`, , zero credit if it does not)
- Direct link to the file on Render.com Prod (Grab the qa url you're testing, paste it, change "-qa" to "-prod"), zero credit if

URL #1
https://github.com/kurtjcastro/kc784-it202-008-2026/M2-Homework/public_html/M2/scenario3.php

URL
https://github.com/kurtjcastro/kc

URL #2
https://kc784-it202-008-prod.onrender.com/M2/scenario3.php

URL
https://kc784-it202-008-prod.onre

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Part 3:

Progress: 100%

Details:
Briefly explain how your solution works. Describe the logic, steps, or decisions your code makes to reach the result. Do not restate the problem or list requirements.

Your Response:

Like scenario 2, I iterated through each array and kept the indexes of the current values, then converted each numeric value to absolute value form using abs() to make them positive. Then I checked their original data types and cast their positive values back to that type before storing them. As for the string values, I checked if their original value had a decimal point so the final output shows either an integer or a decimal looking string value.

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Section #4: (2 pts.) Scenario 4 - Strings

Progress: 100%

☰ Task #1 (2 pts.) - Edit the `transformText` function to solve the challenges

Progress: 100%

Details:

- Only make edits where noted via provided comments
- Challenges will be applied collectively
- Challenge 1: Remove non-alphanumeric characters except spaces
- Challenge 2: Convert text to Title Case
- Challenge 3: Remove leading/trailing spaces and remove duplicate spaces between words
- Result 1-3: Assign final phrase to `placeholderForModifiedPhrase`
- Step 1: sketch out plan using comments (include ucid and date)
- Step 2: Add/commit your outline of comments (required for full credit)
- Step 3: Add code to solve the problem (add/commit as needed)

🖼 Part 1:

Progress: 100%

Details:
Two screenshots are expected

1. Snippet of relevant code showing solution (with ucid/date comment)
2. Full output of executing the program (visit the proper file on Render.com qa after a manual deploy)



relevant code



program execution via render.com qa

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Part 2:

Progress: 100%

Details:

- Direct link to the file in the homework related branch from Github (should end in `.php`, , zero credit if it does not)
- Direct link to the file on Render.com Prod (visit the proper file on Render.com qa after a manual deploy)

URL #1
https://github.com/kurtjcastro/kc784-it202-008-2026/M2-Homework/public_html/M2/scenario4.php

URL
https://github.com/kurtjcastro/kc

URL #2
https://kc784-it202-008-prod.onrender.com/M2/scenario4.php

URL
https://kc784-it202-008-prod.onre

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Part 3:

Progress: 100%

Details:
Briefly explain how your solution works. Describe the logic, steps, or decisions your code makes to reach the result. Do not restate the problem or list requirements.

Your Response:

My code loops through each element in every array and builds a string that only takes alphanumeric characters and spaces. The result is then trimmed down and rechecked by tracking whether the previous character was a space or alphanumeric value. Afterwards, the phrase is converted to title-case by setting characters to lowercase then using ucwords to capitalize the first character of every word.

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Task #2 (+ 1 pt.) - Edit the `transformText` function to solve the extra credit challenge (challenge 4, Extra Credit)

Progress: 100%

Details:

- Only make edits where noted via provided comments

- Challenge 4: Extract up to middle 3 characters (beginning starts at middle of phrase, exclude the first and last character for shorter phrases)
- Assign result to 'placeholderForMiddleCharacters'
- If not enough characters in a word, instead assign "Not enough characters" to `placeholderForMiddleCharacters`
- Step 1: sketch out plan using comments (include ucid and date)
- Step 2: Add/commit your outline of comments (required for full credit)
- Step 3: Add code to solve the problem (add/commit as needed)
- Note: There isn't a URL entry for this since it's part of Scenario 4

Part 1:

Progress: 100%

Details:
Two screenshots are expected

1. Snippet of relevant code showing solution (with ucid/date comment)
2. Full output of executing the program (visit the proper file on Render.com qa after a manual deploy)



relevant code



program execution via render.com qa

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Part 2:

Progress: 100%

Details:

Briefly explain how your solution works. Describe the logic, steps, or decisions your code makes to reach the result. Do not restate the problem or list requirements.

Your Response:

To get the middle 3 characters, I started by finding the starting point by referencing the position closest to the center of the final result, trying my best to avoid including either a starting or ending letter, then used substr to take and store the middle 3 characters.

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Section #5: (2 pts.) Misc

Progress: 100%

Task #1 (0.67 pts.) - Github Details

Progress: 100%

Part 1:

Progress: 100%

Details:
From the Commits tab of the Pull Request, screenshot the commit history Following minimum should be present: addition of baseline files, comments planning thoughts, solutions



commit history

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Part 2:

Progress: 100%

Details:
Include the link to the Pull Request (should end in `/pull/#`, zero credit if it does not)

URL #1
https://github.com/kurtjcastro/kc784-

URL
https://github.com/kurtjcastro/kc

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Task #2 (0.67 pts.) - WakaTime - Activity

Progress: 100%

Details:

- Visit the WakaTime.com Dashboard
- Click `Projects` and find your repository
- Capture the overall time at the top that includes the repository name
- Capture the individual time at the bottom that includes the file time
- Note: The duration isn't relevant for the grade and the visual graphs aren't necessary



overall time



individual time

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Task #3 (0.67 pts.) - Reflection

Progress: 100%

Task #1 (0.33 pts.) - What did you learn?

Progress: 100%

Details:
Briefly answer the question with at least a few reasonable sentences

Your Response:

I learned how to better plan out my approach toward a problem by coding in portions rather than trying to get perfect code on the first try, especially when trying to extract the middle 3 characters in scenario 4. I also touched up on my understanding of loops, casting, and output formatting with different data types.

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Task #2 (0.33 pts.) - What was the easiest part of the assignment?

Progress: 100%

Details:
Briefly answer the question with at least a few reasonable sentences

Your Response:

The easiest part of this assignment was definitely scenarios 1 and 2 because they were both straightforward problems, where scenario 1 was simple math logic, and scenario 2 was just a loop and basic formatting. Both of these problems were pretty much review from prior code-based classes and helped me remember the basic code logic and formatting practice.

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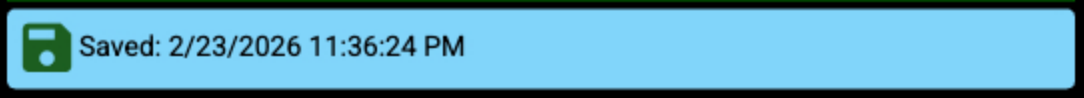
Task #3 (0.33 pts.) - What was the hardest part of the assignment?

Progress: 100%

Details:
Briefly answer the question with at least a few reasonable sentences

Your Response:

The hardest part was definitely scenario 4 because of the amount of data type variety and string rules. Especially during the 3 middle character extraction, oit took a lot more time to figure out how to manipulate the output so that only the middle characters are taken out and properly avoid including the starting or ending characters.


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